

Unit Conversions

The symbols

LePhan Publishing

Contents

1	Acceleration	1
2	Angle	2
3	Angular Velocity	3
4	Area	4
5	Density	5
6	Energy	7
7	Energy Divided By Mass	8
8	Energy Divided By Mole	9
9	Energy Divided By Volume	11
10	Entropy	13
11	Flow Rate (mass basis)	14
12	Flow Rate (mol basis)	16
13	Flow Rate	17
14	Force	19
15	Frequency	20
16	Heat Capacity	13
17	Heat Density	21

18 Heat Flux	23
19 Heat Rate	25
20 Heat Transfer Coefficient	27
21 Length	28
22 Mass	29
23 Mass Divided By Area	30
24 Mass Divided By Length	31
25 Mass Flux	32
26 Mole	34
27 Moment of Inertia	35
28 Momentum	36
29 Pressure	37
30 Pressure Drop	39
31 Specific Heat	41
32 Specific Volume	42
33 Surface Tension	43
34 Temperature	44
35 Thermal Conductivity	45
36 Thermal Diffusivity	47
37 Thermal Expansion (linear)	48
38 Thermal Expansion (volume)	49
39 Thermal Insulance	50

40 Thermal Resistance	52
41 Thermal Resistivity	53
42 Time	55
43 Torque	56
44 Velocity	57
45 Viscosity (dynamic)	59
46 Viscosity (Kinematic)	61
47 Volume	62
48 Volume Divided by Energy	64
49 Work	

Acceleration

Symbol	Name
mm/s^2	millimeter per second squared
mm/min^2	millimeter per minute squared
mm/h^2	millimeter per hour squared
cm/s^2	centimeter per second squared
cm/min^2	centimeter per minute squared
cm/h^2	centimeter per hour squared
m/s^2	meter per second squared
m/min^2	meter per minute squared
m/h^2	meter per hour squared
km/s^2	kilometer per second squared
km/h^2	kilometer per hour squared
Gal	Gal
in/s^2	inch per second squared
in/min^2	inch per minute squared
in/h^2	inch per hour squared
ft/s^2	foot per second squared
ft/min^2	foot per minute squared
ft/h^2	foot per hour squared
yd/s^2	yard per second squared
yd/min^2	yard per minute squared
yd/h^2	yard per hour squared
mi/s^2	mile per second squared
mi/min^2	mile per minute squared
mi/h^2	mile per hour squared

Angle

Symbol	Name
rad	radian
degree	degree
half of circle	half of revolution
r	revolution
second	second
minute	minute
grade	grade
mil	mil

Angular Velocity

Symbol	Name
rad/s	radian per second
rad/min	radian per minute
rad/h	radian per hour
degree per second	degree per second
degree per minute	degree per minute
degree per hour	degree per hour
revolution per second	revolution per second
rpm	revolution per minute
revolution per hour	revolution per hour

Area

Symbol	Name
mm^2	square millimeter
cm^2	square centimeter
m^2	square meter
km^2	square kilometer
in^2	square inch
ft^2	square foot
yd^2	square yard
mi^2	square mile
acre	acre
a	are
b	barn
square chain	square chain
ha	hectare
circular mil	circular mil
square rod	square rod
square micrometer	square micrometer

Density

Symbol	Name
mg/mm^3	milligram per cubic millimeter
mg/cm^3	milligram per cubic centimeter
mg/L	milligram per liter
mg/m^3	milligram per cubic meter
g/mm^3	gram per cubic millimeter
g/cm^3	gram per cubic centimeter
g/L	gram per liter
g/m^3	gram per cubic meter
kg/mm^3	kilogram per cubic millimeter
kg/cm^3	kilogram per cubic centimeter
kg/L	kilogram per liter
kg/m^3	kilogram per cubic meter
metric ton/m^3	metric ton per cubic meter
grain/U.S. gal	grain per U.S. gallon
grain/U.K. gal	grain per U.K. gallon
slug/ft^3	slug per cubic foot

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Density (cont.)

Symbol	Name
avoirdupois ounce/in ³	avoirdupois ounce per cubic inch
avoirdupois ounce/ft ³	avoirdupois ounce per cubic foot
avoirdupois ounce/U.S. gal	avoirdupois ounce per U.S. gallon
avoirdupois ounce/U.K. gal	avoirdupois ounce per U.K. gallon
troy ounce/in ³	troy ounce per cubic inch
troy ounce/ft ³	troy ounce per cubic foot
troy ounce/U.S. gal	troy ounce per U.S. gallon
lb/in ³	pound per cubic inch
lb/ft ³	pound per cubic foot
lb/yd ³	pound per cubic yard
lb/U.S. gal	pound per U.S. gallon
lb/U.K. gal	pound per U.K. gallon
short ton/ft ³	short ton per cubic foot
short ton/yd ³	short ton per cubic yard
long ton/ft ³	long ton per cubic foot
long ton/yd ³	long ton per cubic yard

Energy and Work

Symbol	Name
Btu	British thermal unit
cal	calorie
kcal	kilocalorie
eV	electronvolt
erg	erg
millijoule	millijoule
J	joule
kJ	kilojoule
MJ	megajoule
GJ	gigajoule
quad	quad
ft.poundal	foot poundal
ft.lbf	foot pound force
W.s	watt second
kW.s	kilowatt second
W.h	watt hour
kW.h	kilowatt hour
MW	megawatt hour
GW.h	gigawatt hour
E.C. therm	E.C. therm
U.S. therm	U.S. therm
ton of TNT	ton of TNT

Energy Divided By Mass

Symbol	Name
Btu/lb	British thermal unit per pound
erg/g	erg per gram
J/g	joule per gram
J/kg	joule per kilogram
kJ/kg	kilojoule per kilogram
MJ/kg	megajoule per kilogram
cal/g	calorie per gram
cal/kg	calorie per kilogram
kcal/g	kilocalorie per gram
kcal/kg	kilocalorie per kilogram
cal/lb	calorie per pound
kcal/lb	kilocalorie per pound

Energy Divided By Mole

Symbol	Name
cal/mL	calorie per milliliter
cal/L	calorie per liter
cal/m	calorie per cubic meter
cal/mol	calorie per mole
cal/kmol	calorie per kilomole
cal/lbmol	calorie per poundmole
kcal/mL	kilocalorie per milliliter
kcal/L	kilocalorie per liter
kcal/m ³	kilocalorie per cubic meter
kcal/mol	kilocalorie per mole
kcal/kmol	kilocalorie per kilomole
kcal/lbmol	kilocalorie per poundmole
J/mol	joule per mole
J/kmol	joule per kilomole

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Energy Divided By Mole (cont.)

Symbol	Name
J/mL	joule per milliliter
J/L	joule per liter
J/m ³	joule per cubic meter
kJ/mL	kilojoule per milliliter
kJ/L	kilojoule per liter
kJ/m ³	kilojoule per cubic meter
kJ/mol	kilojoule per mole
kJ/kmol	kilojoule per kilomole
mJ/mol	megajoule per mole
mJ/kmol	megajoule per kilomole
erg/mol	erg per mole
Btu/in ³	British thermal unit per cubic inch
Btu/ft ³	British thermal unit per cubic foot
Btu/lbmol	British thermal unit per poundmole

Energy Divided By Volume

Symbol	Name
Btu/in ³	British thermal unit per cubic inch
Btu/ft ³	British thermal unit per cubic foot
Btu/U.S. gal	British thermal unit per U.S. gallon
Btu/U.K. gal	British thermal unit per U.K. gallon
cal/mm ³	calorie per cubic millimeter
cal/cm ³	calorie per cubic centimeter
cal/mL	calorie per milliliter
cal/L	calorie per liter
cal/m ³	calorie per cubic meter
cal/U.S. gal	calorie per U.S. gallon
cal/U.K. gal	calorie per U.K. gallon
kcal/U.S. gal	kilocalorie per U.S. gallon
kcal/U.K. gal	kilocalorie per U.K. gallon
kcal/cm ³	kilocalorie per cubic centimeter
kcal/mL	kilocalorie per milliliter

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Energy Divided By Volume (cont.)

Symbol	Name
kcal/L	kilocalorie per liter
kcal/m ³	kilocalorie per cubic meter
J/mm ³	joule per cubic millimeter
J/cm ³	joule per cubic centimeter
J/mL	joule per milliliter
J/L	joule per liter
J/m ³	joule per cubic meter
kJ/cm ³	kilojoule per cubic centimeter
kJ/L	kilojoule per liter
kJ/m ³	kilojoule per cubic meter
MJ/m ³	megajoule per cubic meter
W . h/cm ³	watt hour per cubic centimeter
W . h/L	watt hour per liter
W . h/m ³	watt hour per cubic meter
kW . h/m ³	kiowatt hour per cubic meter

Heat Capacity and Entropy

Symbol	Name
Btu/°F	British thermal unit per degree Fahrenheit
Btu/°R	British thermal unit per Rankine
cal/°C	calorie per degree Celsius
cal/°K	calorie per degree Kelvin
kcal/°C	kilocalorie per degree Celsius
kcal/°K	kilocalorie per degree Kelvin
J/°C	joule per degree Celsius
kJ/°C	kilojoule per degree Celsius
MJ/°C	megajoule per degree Celsius
J/°K	joule per degree Kelvin
kJ/°K	kilojoule per degree Kelvin
MJ/°K	megajoule per degree Kelvin

Flow Rate (mass basis)

Symbol	Name
mg/s	milligram per second
mg/min	milligram per minute
mg/h	milligram per hour
g/s	gram per second
g/min	gram per minute
g/h	gram per hour
g/d	gram per day
kg/s	kilogram per second
kg/min	kilogram per minute
kg/h	kilogram per hour
kg/d	kilogram per day
lb/s	pound per second
lb/min	pound per minute
lb/h	pound per hour
lb/d	pound per day

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Flow Rate (mass basis) (cont.)

Symbol	Name
avoirdupois ounce/s	avoirdupois ounce per second
avoirdupois ounce/min	avoirdupois ounce per minute
avoirdupois ounce/h	avoirdupois ounce per hour
troy ounce/s	troy ounce per second
troy ounce/min	troy ounce per minute
troy ounce/h	troy ounce per hour
metric ton/h	metric ton per hour
metric ton/d	metric ton per day
metric ton/year	metric ton per year
short ton/h	short ton per hour
short ton/d	short ton per day
short ton/year	short ton per year
long ton/h	long ton per hour
long ton/d	long ton per day
long ton/year	long ton per year

Flow Rate (mol basis)

Symbol	Name
mL/s	milliliter per second
mL/min	milliliter per minute
mL/h	milliliter per hour
L/s	liter per second
L/min	liter per minute
L/h	liter per hour
m ³ /s	cubic meter per second
m ³ /min	cubic meter per minute
m ³ /h	cubic meter per hour
mol/s	mol per second
mol/min	mol per minute
mol/h	mol per hour
kmol/s	kilo mol per second
kmol/min	kilo mol per minute
kmol/h	kilo mol per hour
lbmol/s	pound mol per second
lbmol/min	pound mol per minute
lbmol/h	pound mol per hour
in ³ /s	cubic inch per second
in ³ /min	cubic inch per minute
in ³ /h	cubic inch per hour
ft ³ /s	cubic foot per second
ft ³ /min	cubic foot per minute
ft ³ /h	cubic foot per hour

Flow Rate

Symbol	Name
U.S. petroleum barrel/h	U.S.petroleum barrel per hour
U.S. petroleum barrel/d	U.S.petroleum barrel per day
U.S. petroleum barrel/year	U.S.petroleum barrel per year
U.K. barrel/h	U.K. barrel per hour
U.K. barrel/d	U.K. barrel per day
U.K. barrel/year	U.K. barrel per year
mm^3/s	cubic millimeter per second
mm^3/min	cubic millimeter per minute
cm^3/s	cubic centimeter per second
cm^3/min	cubic centimeter per minute
cm^3/h	cubic centimeter per hour
mL/s	milliliter per second
mL/min	milliliter per minute
mL/h	milliliter per hour
L/s	liter per second
L/min	liter per minute
L/h	liter per hour
L/d	liter per day
m^3/s	cubic meter per second
m^3/min	cubic meter per minute
m^3/h	cubic meter per hour
m^3/d	cubic meter per day
m^3/year	cubic meter per year

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Flow Rate (continue)

Symbol	Name
in^3/s	cubic inch per second
in^3/min	cubic inch per minute
in^3/h	cubic inch per hour
ft^3/s	cubic foot per second
ft^3/min	cubic foot per minute
ft^3/h	cubic foot per hour
ft^3/day	cubic foot per day
U.S. gal/s	U.S. gallon per second
U.S. gal/min	U.S. gallon per minute
U.S. gal/h	U.S. gallon per hour
U.S. gal/d	U.S. gallon per day
U.S. gal/year	U.S. gallon per year
U.K. gal/s	U.K. gallon per second
U.K. gal/min	U.K. gallon per minute
U.K. gal/h	U.K. gallon per hour
U.K. gal/d	U.K. gallon per day
U.K. gal/year	U.K. gallon per year

Force

Symbol	Name
avoirdupois ounce-force	avoirdupois ounce-force
troy ounce-force	troy ounce-force
dyn	dyne
grain-force	grain-force
mN	millinewton
N	newton
kN	kilonewton
MN	meganewton
lbf	pound-force
poundal	poundal
gf	gram-force
kgf	kilogram-force
kip	kip
metric ton-force	metric ton-force
short ton-force	short ton-force
long ton-force	long ton-force

Frequency

Symbol	Name
Hz	hertz
kHz	kilohertz
MHz	megahertz
1/s	inversion of second
1/min	inversion of minute
1/h	inversion of hour
1/d	inversion of day

Heat Density

Symbol	Name
Btu/in ²	British thermal unit per square inch
Btu/ft ²	British thermal unit per square foot
Btu/yd ²	British thermal unit per square yard
Btu/mi ²	British thermal unit per square mile
million Btu/mi ²	million British thermal unit per square mile
cal/mm ²	calorie per square millimeter
cal/cm ²	calorie per square centimeter
cal/m ²	calorie per square meter
cal/in ²	calorie per square inch
cal/ft ²	calorie per square foot
kcal/cm ²	kilocalorie per square centimeter
kcal/m ²	kilocalorie per square meter
kcal/in ²	kilocalorie per square inch
kcal/ft ²	kilocalorie per square foot

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Heat Density (cont.)

Symbol	Name
J/mm^2	joule per square millimeter
J/cm^2	joule per square centimeter
J/m^2	joule per square meter
kJ/cm^2	kilojoule per square centimeter
kJ/m^2	kilojoule per square meter
MJ/m^2	megajoule per square meter
MJ/km^2	megajoule per square kilometer
$\text{W} \cdot \text{h/mm}^2$	watt hour per square millimeter
$\text{W} \cdot \text{h/cm}^2$	watt hour per square centimeter
$\text{W} \cdot \text{h/m}^2$	watt hour per square meter
$\text{kW} \cdot \text{h/m}^2$	kilowatt hour per square meter
$\text{kW} \cdot \text{h/km}^2$	kilowatt hour per square kilometer
$\text{MW} \cdot \text{h/m}^2$	megawatt hour per square meter
$\text{MW} \cdot \text{h/km}^2$	megawatt hour per square kilometer

Heat Flux

Symbol	Name
$\text{Btu}/(\text{s} \cdot \text{in}^2)$	British thermal unit per second square inch
$\text{Btu}/(\text{min} \cdot \text{in}^2)$	British thermal unit per minute square inch
$\text{Btu}/(\text{h} \cdot \text{in}^2)$	British thermal unit per hour square inch
$\text{Btu}/(\text{s} \cdot \text{ft}^2)$	British thermal unit per second square foot
$\text{Btu}/(\text{min} \cdot \text{ft}^2)$	British thermal unit per minute square foot
$\text{Btu}/(\text{d} \cdot \text{ft}^2)$	British thermal unit per day square foot
$\text{Btu}/(\text{h} \cdot \text{ft}^2)$	British thermal unit per hour square foot
million $\text{Btu}/(\text{h} \cdot \text{ft}^2)$	million British thermal unit per hour square foot
million $\text{Btu}/(\text{d} \cdot \text{ft}^2)$	million British thermal unit per day square foot
$\text{J}/(\text{h} \cdot \text{mm}^2)$	joule per hour square millimeter
$\text{J}/(\text{h} \cdot \text{cm}^2)$	joule per hour square centimeter
$\text{J}/(\text{h} \cdot \text{m}^2)$	joule per hour square meter
$\text{kJ}/(\text{h} \cdot \text{m}^2)$	kilojoule per hour square meter
$\text{kJ}/(\text{d} \cdot \text{m}^2)$	kilojoule per day square meter
$\text{MJ}/(\text{h} \cdot \text{m}^2)$	megajoule per hour square meter
$\text{MJ}/(\text{d} \cdot \text{m}^2)$	megajoule per day square meter

(... continue in next page)

Heat Flux (cont.)

Symbol	Name
$\text{cal}/(\text{s} \cdot \text{mm}^2)$	calorie per second square millimeter
$\text{cal}/(\text{s} \cdot \text{cm}^2)$	calorie per second square centimeter
$\text{cal}/(\text{min} \cdot \text{cm}^2)$	calorie per minute square centimeter
$\text{cal}/(\text{h} \cdot \text{m}^2)$	calorie per hour square centimeter
$\text{kcal}/(\text{s} \cdot \text{cm}^2)$	kilocalorie per second square centimeter
$\text{kcal}/(\text{min} \cdot \text{cm}^2)$	kilocalorie per minute square centimeter
$\text{kcal}/(\text{h} \cdot \text{cm}^2)$	kilocalorie per hour square centimeter
$\text{kcal}/(\text{d} \cdot \text{cm}^2)$	kilocalorie per day square centimeter
$\text{kcal}/(\text{s} \cdot \text{m}^2)$	kilocalorie per second square meter
$\text{kcal}/(\text{h} \cdot \text{m}^2)$	kilocalorie per hour square meter
$\text{kcal}/(\text{d} \cdot \text{m}^2)$	kilocalorie per day square meter
W/mm^2	watt per square millimeter
W/cm^2	watt per square centimeter
W/m^2	watt per square meter
kW/m^2	kilowatt per square meter
MW/m^2	megawatt per square meter

Heat Rate

Symbol	Name
Btu/s	British thermal unit per second
Btu/min	British thermal unit per minute
Btu/h	British thermal unit per hour
Btu/d	British thermal unit per day
million Btu/h	million British thermal unit per hour
million Btu/d	million British thermal unit per day
cal/s	calorie per second
cal/min	calorie per minute
cal/h	calorie per hour
kcal/s	kilocalorie per second
kcal/min	kilocalorie per minute
kcal/h	kilocalorie per hour
kcal/d	kilocalorie per day
erg/s	erg per second

(... continue in next page)

Heat Rate (cont.)

Symbol	Name
J/h	joule per hour
kJ/h	kilojoule per hour
kJ/d	kilojoule per day
MJ/h	megajoule per hour
MJ/d	megajoule per day
ft . lbf/s	foot pound-force per second
ft . lbf/min	foot pound-force per minute
ft . lbf/h	foot pound-force per hour
metric horsepower	metric horsepower
ton of refrigeration	ton of refrigeration
W	watt
kW	kilowatt
MW	megawatt
GW	gigawatt

Heat Transfer Coefficient

Symbol	Name
$\text{Btu}/(\text{s} \cdot \text{in}^2 \cdot ^\circ\text{F})$	British thermal unit per second square inch degree Fahrenheit
$\text{Btu}/(\text{h} \cdot \text{in}^2 \cdot ^\circ\text{F})$	British thermal unit per hour square inch degree Fahrenheit
$\text{Btu}/(\text{s} \cdot \text{ft}^2 \cdot ^\circ\text{F})$	British thermal unit per second square foot degree Fahrenheit
$\text{Btu}/(\text{h} \cdot \text{ft}^2 \cdot ^\circ\text{F})$	British thermal unit per hour square foot degree Fahrenheit
$\text{Btu}/(\text{d} \cdot \text{ft}^2 \cdot ^\circ\text{F})$	British thermal unit per day square foot degree Fahrenheit
$\text{cal}/(\text{s} \cdot \text{mm}^2 \cdot ^\circ\text{C})$	calorie per second square millimeter degree Celsius
$\text{cal}/(\text{s} \cdot \text{cm}^2 \cdot ^\circ\text{C})$	calorie per second square centimeter degree Celsius
$\text{cal}/(\text{h} \cdot \text{cm}^2 \cdot ^\circ\text{C})$	calorie per hour square centimeter degree Celsius
$\text{cal}/(\text{h} \cdot \text{m}^2 \cdot ^\circ\text{C})$	calorie per hour square meter degree Celsius
$\text{kcal}/(\text{s} \cdot \text{m}^2 \cdot ^\circ\text{C})$	kilcalorie per second square meter degree Celsius
$\text{kcal}/(\text{h} \cdot \text{m}^2 \cdot ^\circ\text{C})$	kilcalorie per hour square meter degree Celsius
$\text{W}/(\text{mm}^2 \cdot ^\circ\text{C})$	watt per square millimeter degree Celsius
$\text{W}/(\text{cm}^2 \cdot ^\circ\text{C})$	watt per square centimeter degree Celsius
$\text{W}/(\text{m}^2 \cdot ^\circ\text{C})$	watt per square meter degree Celsius
$\text{kW}/(\text{m}^2 \cdot ^\circ\text{C})$	kilowatt per square meter degree Celsius
$\text{MW}/(\text{m}^2 \cdot ^\circ\text{C})$	megawatt per square meter degree Celsius
$\text{kJ}/(\text{h} \cdot \text{m}^2 \cdot ^\circ\text{C})$	kilojoule per hour square meter degree Celsius
$\text{kJ}/(\text{h} \cdot \text{m}^2 \cdot ^\circ\text{K})$	kilojoule per hour square meter degree Kelvin
$\text{MJ}/(\text{h} \cdot \text{m}^2 \cdot ^\circ\text{K})$	megajoule per hour square meter degree Kelvin
$\text{W}/(\text{m}^2 \cdot ^\circ\text{K})$	watt per square meter degree Kelvin
$\text{kW}/(\text{m}^2 \cdot ^\circ\text{K})$	kilowatt per square meter degree Kelvin
$\text{MW}/(\text{m}^2 \cdot ^\circ\text{K})$	megawatt per square meter degree Kelvin

Length

Symbol	Name
mm	millimeter
cm	centimeter
m	meter
km	kilometer
in	inch
yd	yard
ft	foot
mi	mile
angstrom	angstrom
nanometer	nanometer
micron	micron
micrometer	micrometer
chain	U.S. survey foot chain
computer point	computer point
mil	mil
rod	rod

Mass

Symbol	Name
mg	milligram
g	gram
kg	kilogram
carat	carat
grain	grain
oz	avoirdupois ounce
oz	troy ounce
lb	pound
slug	slug
stone	stone
t	metric ton
short ton	short ton
long ton	long ton
short hundredweight	short hundredweight
long hundredweight	long hundredweight
dwt	pennyweight

Mass Divided By Area

Symbol	Name
g/mm^2	gram per square millimeter
g/cm^2	gram per square centimeter
g/m^2	gram per square meter
kg/mm^2	kilogram per square millimeter
kg/cm^2	kilogram per square centimeter
kg/m^2	kilogram per square meter
kg/km^2	kilogram per square kilometer
metric ton/m^2	metric ton per square meter
metric ton/km^2	metric ton per square kilometer
short ton/ft^2	short ton per square foot
short ton/yd^2	short ton per square yard
short ton/mi^2	short ton per square mile
$\text{avoirdupois ounce/in}^2$	avoirdupois ounce per square inch
$\text{avoirdupois ounce/ft}^2$	avoirdupois ounce per square foot
$\text{avoirdupois ounce/yd}^2$	avoirdupois ounce per square yard
lb/in^2	pound per square inch
lb/ft^2	pound per square foot
lb/yd^2	pound per square yard
lb/mi^2	pound per square mile
slug/in^2	slug per square inch
slug/ft^2	slug per square foot
long ton/ft^2	long ton per square foot
long ton/yd^2	long ton per square yard
long ton/mi^2	long ton per square mile

Mass Divided By Length

Symbol	Name
g/mm	gram per millimeter
g/cm	gram per centimeter
g/m	gram per meter
kg/mm	kilogram per millimeter
kg/cm	kilogram per centimeter
kg/m	kilogram per meter
kg/km	kilogram per kilometer
metric ton/m	metric ton per meter
metric ton/km	metric ton per kilometer
short ton/ft	short ton per foot
short ton/yd	short ton per yard
short ton/mi	short ton per mile
avoirdupois ounce/in	avoirdupois ounce per inch
avoirdupois ounce/ft	avoirdupois ounce per foot
avoirdupois ounce/yd	avoirdupois ounce per yard
lb/in	pound per inch
lb/ft	pound per foot
lb/yd	pound per yard
lb/mi	pound per mile
slug/in	slug per inch
slug/ft	slug per foot
long ton/ft	long ton per foot
long ton/yd	long ton per yard
long ton/mi	long ton per mile

Mass Flux

Symbol	Name
$\text{g}/(\text{s} \cdot \text{mm}^2)$	gram per second square millimeter
$\text{g}/(\text{s} \cdot \text{cm}^2)$	gram per second square centimeter
$\text{g}/(\text{s} \cdot \text{m}^2)$	gram per second square meter
$\text{g}/(\text{min} \cdot \text{mm}^2)$	gram per minute square millimeter
$\text{g}/(\text{min} \cdot \text{cm}^2)$	gram per minute square centimeter
$\text{g}/(\text{min} \cdot \text{m}^2)$	gram per minute square meter
$\text{g}/(\text{h} \cdot \text{mm}^2)$	gram per hour square millimeter
$\text{g}/(\text{h} \cdot \text{cm}^2)$	gram per hour square centimeter
$\text{g}/(\text{h} \cdot \text{m}^2)$	gram per hour square meter
$\text{kg}/(\text{s} \cdot \text{mm}^2)$	kilogram per second square millimeter
$\text{kg}/(\text{s} \cdot \text{cm}^2)$	kilogram per second square centimeter
$\text{kg}/(\text{s} \cdot \text{m}^2)$	kilogram per second square meter
$\text{kg}/(\text{min} \cdot \text{mm}^2)$	kilogram per minute square millimeter
$\text{kg}/(\text{min} \cdot \text{cm}^2)$	kilogram per minute square centimeter
$\text{kg}/(\text{min} \cdot \text{m}^2)$	kilogram per minute square meter
$\text{kg}/(\text{h} \cdot \text{mm}^2)$	kilogram per hour square millimeter
$\text{kg}/(\text{h} \cdot \text{cm}^2)$	kilogram per hour square centimeter

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Mass Flux (cont.)

Symbol	Name
$\text{kg}/(\text{h} \cdot \text{m}^2)$	kilogram per hour square meter
$\text{kg}/(\text{h} \cdot \text{km}^2)$	kilogram per hour square kilometer
$\text{metric ton}/(\text{h} \cdot \text{m}^2)$	metric ton per hour square meter
$\text{metric ton}/(\text{h} \cdot \text{km}^2)$	metric ton per hour square kilometer
$\text{lb}/(\text{s} \cdot \text{in}^2)$	pound per second square inch
$\text{lb}/(\text{s} \cdot \text{ft}^2)$	pound per second square foot
$\text{lb}/(\text{s} \cdot \text{yd}^2)$	pound per second square yard
$\text{lb}/(\text{min} \cdot \text{in}^2)$	pound per minute square inch
$\text{lb}/(\text{min} \cdot \text{ft}^2)$	pound per minute square foot
$\text{lb}/(\text{min} \cdot \text{yd}^2)$	pound per minute square yard
$\text{lb}/(\text{h} \cdot \text{in}^2)$	pound per hour square inch
$\text{lb}/(\text{h} \cdot \text{ft}^2)$	pound per hour square foot
$\text{lb}/(\text{h} \cdot \text{yd}^2)$	pound per hour square yard
$\text{short ton}/(\text{h} \cdot \text{yd}^2)$	short ton per hour square yard
$\text{short ton}/(\text{h} \cdot \text{mi}^2)$	short ton per hour square mile
$\text{long ton}/(\text{h} \cdot \text{yd}^2)$	long ton per hour square yard
$\text{long ton}/(\text{h} \cdot \text{mi}^2)$	long ton per hour square mile

Mole

Symbol	Name
mL	milliliter
L	liter
m ³	cubic meter
mol	mol
kmol	kilo mol
lbmol	pound mol
in ³	cubic inch
ft ³	cubic foot

Moment of Inertia

Symbol	Name
$\text{g} \cdot \text{mm}^2$	gram square millimeter
$\text{g} \cdot \text{cm}^2$	gram square centimeter
$\text{g} \cdot \text{m}^2$	gram square meter
$\text{kg} \cdot \text{mm}^2$	kilogram square millimeter
$\text{kg} \cdot \text{cm}^2$	kilogram square centimeter
$\text{kg} \cdot \text{m}^2$	kilogram square meter
metric ton $\cdot \text{m}^2$	metric ton square meter
tonne $\cdot \text{m}^2$	tonne square meter
avoirdupois ounce $\cdot \text{in}^2$	avoirdupois ounce square inch
avoirdupois ounce $\cdot \text{ft}^2$	avoirdupois ounce square foot
$\text{lb} \cdot \text{in}^2$	pound square inch
$\text{lb} \cdot \text{ft}^2$	pound square foot
$\text{slug} \cdot \text{in}^2$	slug square inch
$\text{slug} \cdot \text{ft}^2$	slug square foot
short ton $\cdot \text{ft}^2$	short ton square foot
long ton $\cdot \text{ft}^2$	long ton square foot

Momentum

Symbol	Name
g . mm/s	gram millimeter per second
g . cm/s	gram centimeter per second
kg . cm/s	kilogram centimeter per second
kg . m/s	kilogram meter per second
kg . cm/h	kilogram centimeter per hour
kg . m/h	kilogram meter per hour
lb . in/s	pound inch per second
lb . in/h	pound inch per hour
lb . ft/s	pound foot per second
lb . ft/h	pound foot per hour

Pressure

Symbol	Name
atm	atmosphere
bar	bar
millibar	millibar
dyn/cm ²	dyne per square centimeter
gf/mm ²	gram-force per square millimeter
gf/m ²	gram-force per square meter
gf/cm ²	gram-force per square centimeter
kgf/mm ²	kilogram-force per square millimeter
kgf/cm ²	kilogram-force per square centimeter
kgf/m ²	kilogram-force per square meter
mm Hg	millimeter of mercury
cm Hg	centimeter of mercury
m Hg	meter of mercury
in Hg	inch of mercury
ft Hg	foot of mercury
kip/in ²	kip per square inch
kip/ft ²	kip per square foot

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Pressure (cont.)

Symbol	Name
N/mm^2	newton per square millimeter
N/cm^2	newton per square centimeter
N/m^2	newton per square meter
Pa	pascal
kPa	kilopascal
MPa	megapascal
psi	psi
lbf/in^2	pound-force per square inch
lbf/ft^2	pound-force per square foot
lbf/in^2	poundal per square inch
lbf/ft^2	poundal per square foot
torr	torr
mm H ₂ O	millimeter of water
cm H ₂ O	centimeter of water
m H ₂ O	meter of water
in H ₂ O	inch of water
ft H ₂ O	foot of water

Pressure Drop

Symbol	Name
mm H ₂ O/mm	millimeter of water per millimeter
mm H ₂ O/cm	millimeter of water per centimeter
mm H ₂ O/m	millimeter of water per meter
cm H ₂ O/mm	centimeter of water per millimeter
cm H ₂ O/cm	centimeter of water per centimeter
cm H ₂ O/m	centimeter of water per meter
m H ₂ O/mm	meter of water per millimeter
m H ₂ O/cm	meter of water per centimeter
m H ₂ O/m	meter of water per meter
mm Hg/mm	millimeter of mercury per millimeter
mm Hg/cm	millimeter of mercury per centimeter
mm Hg/m	millimeter of mercury per meter
cm Hg/mm	centimeter of mercury per millimeter
cm Hg/cm	centimeter of mercury per centimeter
cm Hg/m	centimeter of mercury per meter
Pa/mm	pascal per millimeter
Pa/cm	pascal per centimeter
Pa/m	pascal per meter

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Pressure Drop (cont.)

Symbol	Name
kPa/m	kilopascal per meter
kPa/km	kilopascal per kilometer
MPa/m	megapascal per meter
MPa/km	megapascal per kilometer
$(\text{kgf}/\text{cm}^2) / \text{m}$	kilogram-force per square centimeter per meter
$(\text{kgf}/\text{m}^2) / \text{m}$	kilogram-force per square meter per meter
bar/m	bar per meter
psi/in	pound-force per square inch per inch
psi/ft	pound-force per square inch per foot
psi/yd	pound-force per square inch per yard
psf/in	pound-force per square foot per inch
psf/ft	pound-force per square foot per foot
psf/yd	pound-force per square foot per yard
psf/mi	pound-force per square foot per mile
in H ₂ O/in	inch of water per inch
in H ₂ O/ft	inch of water per foot
ft H ₂ O/in	foot of water per inch
ft H ₂ O/ft	foot of water per foot

Specific Heat

Symbol	Name
Btu/(lb . °F)	British thermal unit per pound Fahrenheit
Btu/(lb . °R)	British thermal unit per pound Rankine
erg/(g . °C)	erg per gram Celsius
erg/(g . °K)	erg per gram Kelvin
J/(g . °C)	joule per gram Celsius
J/(kg . °C)	joule per kilogram Celsius
kJ/(kg . °C)	kilojoule per kilogram Celsius
MJ/(kg . °C)	megajoule per kilogram Celsius
J/(g . °K)	joule per gram Kelvin
J/(kg . °K)	joule per kilogram Kelvin
kJ/(kg . °K)	kilojoule per kilogram Kelvin
MJ/(kg . °K)	megajoule per kilogram Kelvin
cal/(g . °C)	calorie per gram Celsius
cal/(kg . °C)	calorie per kilogram Celsius
cal/(lb . °C)	calorie per pound Celsius
cal/(lb . °F)	calorie per pound Fahrenheit
cal/(g . °K)	calorie per gram Kelvin
cal/(kg . °K)	calorie per kilogram Kelvin
kcal/(g . °C)	kilocalorie per gram Celsius
kcal/(kg . °C)	kilocalorie per kilogram Celsius
kcal/(lb . °C)	kilocalorie per pound Celsius
kcal/(lb . °F)	kilocalorie per pound Fahrenheit
kcal/(g . °K)	kilocalorie per gram Kelvin
kcal/(kg . °K)	kilocalorie per kilogram Kelvin

Specific Volume

Symbol	Name
mm^3/g	cubic millimeter per gram
mm^3/kg	cubic millimeter per kilogram
cm^3/g	cubic centimeter per gram
cm^3/kg	cubic centimeter per kilogram
m^3/kg	cubic meter per kilogram
$\text{m}^3/\text{metric ton}$	cubic meter per metric ton
mL/g	milliliter per gram
mL/kg	milliliter per kilogram
L/g	liter per gram
L/kg	liter per kilogram
in^3/lb	cubic inch per pound
ft^3/lb	cubic foot per pound
U.K. gal/lb	U.K. gallon per pound
U.K. barrel/lb	U.K. barrel per pound
U.K. barrel/short ton	U.K. barrel per short ton
U.K. barrel/long ton	U.K. barrel per long ton
U.S. gal/lb	U.S. gallon per pound
U.S. petroleum barrel/lb	U.S. petroleum barrel per pound
U.S. petroleum barrel/short ton	U.S. petroleum barrel per short ton
U.S. petroleum barrel/long ton	U.S. petroleum barrel per long ton

Surface Tension

Symbol	Name
dyne/cm	dyne per centimeter
mN/mm	millinewton per millimeter
mN/cm	millinewton per centimeter
mN/m	millinewton per meter
N/mm	newton per millimeter
N/cm	newton per centimeter
N/m	newton per meter
kN/mm	kilonewton per millimeter
kN/cm	kilonewton per centimeter
kN/m	kilonewton per meter
MN/m	meganeutron per meter
kgf/mm	kilogram-force per millimeter
kgf/cm	kilogram-force per centimeter
kgf/m	kilogram-force per meter
lbf/in	pound-force per inch
lbf/ft	pound-force per foot
poundal/in	poundal per inch
poundal/ft	poundal per foot
short ton-force/in	short ton-force per inch
short ton-force/ft	short ton-force per foot
long ton-force/in	long ton-force per inch
long ton-force/ft	long ton-force per foot

Temperature

Symbol	Name
°C	degree Celsius
°F	degree Fahrenheit
°K	degree Kelvin
°R	degree Rankine

Thermal Conductivity

Symbol	Name
$\text{Btu}/(\text{s} \cdot \text{in} \cdot ^\circ\text{F})$	British thermal unit per second inch degree Fahrenheit
$\text{Btu}/(\text{s} \cdot \text{ft} \cdot ^\circ\text{F})$	British thermal unit per second foot degree Fahrenheit
$\text{Btu}/(\text{h} \cdot \text{in} \cdot ^\circ\text{F})$	British thermal unit per hour inch degree Fahrenheit
$\text{Btu}/(\text{h} \cdot \text{ft} \cdot ^\circ\text{F})$	British thermal unit per hour foot degree Fahrenheit
$\text{Btu}/(\text{s} \cdot \text{in} \cdot ^\circ\text{R})$	British thermal unit per second inch degree Rankine
$\text{Btu}/(\text{s} \cdot \text{ft} \cdot ^\circ\text{R})$	British thermal unit per second foot degree Rankine
$\text{Btu}/(\text{h} \cdot \text{in} \cdot ^\circ\text{R})$	British thermal unit per hour inch degree Rankine
$\text{Btu}/(\text{h} \cdot \text{ft} \cdot ^\circ\text{R})$	British thermal unit per hour foot degree Rankine
$\text{cal}/(\text{s} \cdot \text{mm} \cdot ^\circ\text{C})$	calorie per second millimeter degree Celsius
$\text{cal}/(\text{s} \cdot \text{cm} \cdot ^\circ\text{C})$	calorie per second centimeter degree Celsius
$\text{cal}/(\text{s} \cdot \text{m} \cdot ^\circ\text{C})$	calorie per second meter degree Celsius
$\text{kcal}/(\text{s} \cdot \text{m} \cdot ^\circ\text{C})$	kilocalorie per second meter degree Celsius
$\text{kcal}/(\text{min} \cdot \text{m} \cdot ^\circ\text{C})$	kilocalorie per minute meter degree Celsius
$\text{kcal}/(\text{h} \cdot \text{m} \cdot ^\circ\text{C})$	kilocalorie per hour meter degree Celsius

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Thermal Conductivity (cont.)

Symbol	Name
$\text{W}/(\text{mm} \cdot ^\circ\text{C})$	watt per millimeter degree Celsius
$\text{W}/(\text{cm} \cdot ^\circ\text{C})$	watt per centimeter degree Celsius
$\text{W}/(\text{m} \cdot ^\circ\text{C})$	watt per meter degree Celsius
$\text{kW}/(\text{m} \cdot ^\circ\text{C})$	kilowatt per meter degree Celsius
$\text{MW}/(\text{m} \cdot ^\circ\text{C})$	megawatt per meter degree Celsius
$\text{kJ}/(\text{h} \cdot \text{m} \cdot ^\circ\text{C})$	kilojoule per hour meter degree Celsius
$\text{MJ}/(\text{h} \cdot \text{m} \cdot ^\circ\text{C})$	megajoule per hour meter degree Celsius
$\text{kJ}/(\text{h} \cdot \text{m} \cdot ^\circ\text{K})$	kilojoule per hour meter degree Kelvin
$\text{MJ}/(\text{h} \cdot \text{m} \cdot ^\circ\text{K})$	megajoule per hour meter degree Kelvin
$\text{W}/(\text{mm} \cdot ^\circ\text{K})$	watt per millimeter degree Kelvin
$\text{W}/(\text{cm} \cdot ^\circ\text{K})$	watt per centimeter degree Kelvin
$\text{W}/(\text{m} \cdot ^\circ\text{K})$	watt per meter degree Kelvin
$\text{kW}/(\text{m} \cdot ^\circ\text{K})$	kilowatt per meter degree Kelvin
$\text{mW}/(\text{m} \cdot ^\circ\text{K})$	megawatt per meter degree Kelvin

Thermal Diffusivity

Symbol	Name
mm^2/s	square millimeter per second
cm^2/s	square centimeter per second
m^2/s	square meter per second
mm^2/h	square millimeter per hour
cm^2/h	square centimeter per hour
m^2/h	square meter per hour
in^2/s	square inch per second
ft^2/s	square foot per second
in^2/h	square inch per hour
ft^2/h	square foot per hour

Thermal Expansion (linear)

Symbol	Name
angstrom/°C	angstrom per degree Celsius
nanometer/°C	nanometer per degree Celsius
micrometer/°C	micrometer per degree Celsius
mm/°C	millimeter per degree Celsius
cm/°C	centimeter per degree Celsius
m/°C	meter per degree Celsius
in/°F	inch per degree Fahrenheit
angstrom/°K	angstrom per degree Kelvin
nanometer/°K	nanometer per degree Kelvin
micrometer/°K	micrometer per degree Kelvin
mm/°K	millimeter per degree Kelvin
cm/°K	centimeter per degree Kelvin
m/°K	meter per degree Kelvin
ft/°F	foot per degree Fahrenheit

Thermal Expansion (volume)

Symbol	Name
$\text{mm}^3 / ^\circ\text{C}$	cubic millimeter per degree Celsius
$\text{cm}^3 / ^\circ\text{C}$	cubic centimeter per degree Celsius
$\text{m}^3 / ^\circ\text{C}$	cubic meter per degree Celsius
$\text{mm}^3 / ^\circ\text{K}$	cubic millimeter per degree Kelvin
$\text{cm}^3 / ^\circ\text{K}$	cubic centimeter per degree Kelvin
$\text{m}^3 / ^\circ\text{K}$	cubic meter per degree Kelvin
U.S petroleum barrel/ $^\circ\text{F}$	U.S petroleum barrel per degree Fahrenheit
$\text{mL} / ^\circ\text{C}$	milliliter per degree Celsius
$\text{L} / ^\circ\text{C}$	liter per degree Celsius
$\text{in}^3 / ^\circ\text{F}$	cubic inch per degree Fahrenheit
$\text{ft}^3 / ^\circ\text{F}$	cubic foot per degree Fahrenheit
$\text{yd}^3 / ^\circ\text{F}$	cubic yard per degree Fahrenheit
U.K. gal/ $^\circ\text{F}$	U.K. gallon per degree Fahrenheit
U.S. gal/ $^\circ\text{F}$	U.S. gallon per degree Fahrenheit

Thermal Insulance

Symbol	Name
$^{\circ}\text{C} \cdot \text{mm}^2/\text{W}$	degree Celsius square millimeter per watt
$^{\circ}\text{C} \cdot \text{cm}^2/\text{W}$	degree Celsius square centimeter per watt
$^{\circ}\text{C} \cdot \text{m}^2/\text{W}$	degree Celsius square meter per watt
$^{\circ}\text{C} \cdot \text{mm}^2/\text{kW}$	degree Celsius square millimeter per kilowatt
$^{\circ}\text{C} \cdot \text{cm}^2/\text{kW}$	degree Celsius square centimeter per kilowatt
$^{\circ}\text{C} \cdot \text{m}^2/\text{kW}$	degree Celsius square meter per kilowatt
$^{\circ}\text{C} \cdot \text{s} \cdot \text{mm}^2/\text{cal}$	degree Celsius second square millimeter per calorie
$^{\circ}\text{C} \cdot \text{s} \cdot \text{cm}^2/\text{cal}$	degree Celsius second square centimeter per calorie
$^{\circ}\text{C} \cdot \text{s} \cdot \text{m}^2/\text{cal}$	degree Celsius second square meter per calorie
$^{\circ}\text{C} \cdot \text{s} \cdot \text{mm}^2/\text{kcal}$	degree Celsius second square millimeter per kilocalorie
$^{\circ}\text{C} \cdot \text{s} \cdot \text{cm}^2/\text{kcal}$	degree Celsius second square centimeter per kilocalorie
$^{\circ}\text{C} \cdot \text{s} \cdot \text{m}^2/\text{kcal}$	degree Celsius second square meter per kilocalorie
$^{\circ}\text{K} \cdot \text{m}^2/\text{W}$	degree Kelvin square meter per watt
$^{\circ}\text{K} \cdot \text{m}^2/\text{kW}$	degree Kelvin square meter per kilowatt
$^{\circ}\text{K} \cdot \text{m}^2/\text{MW}$	degree Kelvin square meter per megawatt

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Thermal Insulance (cont.)

Symbol	Name
$^{\circ}\text{K} \cdot \text{h} \cdot \text{m}^2/\text{J}$	degree Kelvin hour square meter per joule
$^{\circ}\text{K} \cdot \text{h} \cdot \text{m}^2/\text{kJ}$	degree Kelvin hour square meter per kilojoule
$^{\circ}\text{K} \cdot \text{h} \cdot \text{m}^2/\text{MJ}$	degree Kelvin hour square meter per megajoule
$^{\circ}\text{F} \cdot \text{s} \cdot \text{in}^2/\text{Btu}$	degree Fahrenheit second square inch per Btu
$^{\circ}\text{F} \cdot \text{s} \cdot \text{ft}^2/\text{Btu}$	degree Fahrenheit second square foot per Btu
$^{\circ}\text{F} \cdot \text{h} \cdot \text{in}^2/\text{Btu}$	degree Fahrenheit hour square inch per Btu
$^{\circ}\text{F} \cdot \text{h} \cdot \text{ft}^2/\text{Btu}$	degree Fahrenheit hour square foot per Btu
$^{\circ}\text{F} \cdot \text{in}^2/\text{W}$	degree Fahrenheit square inch per watt
$^{\circ}\text{F} \cdot \text{ft}^2/\text{W}$	degree Fahrenheit square foot per watt
$^{\circ}\text{F} \cdot \text{in}^2/\text{kW}$	degree Fahrenheit square inch per kilowatt
$^{\circ}\text{F} \cdot \text{ft}^2/\text{kW}$	degree Fahrenheit square foot per kilowatt
$^{\circ}\text{F} \cdot \text{h} \cdot \text{in}^2/\text{J}$	degree Fahrenheit hour square inch per joule
$^{\circ}\text{F} \cdot \text{h} \cdot \text{ft}^2/\text{J}$	degree Fahrenheit hour square foot per joule
$^{\circ}\text{F} \cdot \text{h} \cdot \text{in}^2/\text{kJ}$	degree Fahrenheit hour square inch per kilojoule
$^{\circ}\text{F} \cdot \text{h} \cdot \text{ft}^2/\text{kJ}$	degree Fahrenheit hour square foot per kilojoule

Thermal Resistance

Symbol	Name
$^{\circ}\text{C}/\text{W}$	degree Celsius per watt
$^{\circ}\text{C}/\text{kW}$	degree Celsius per kilowatt
$^{\circ}\text{C} \cdot \text{s}/\text{cal}$	degree Celsius second per calorie
$^{\circ}\text{C} \cdot \text{min}/\text{cal}$	degree Celsius minute per calorie
$^{\circ}\text{C} \cdot \text{h}/\text{cal}$	degree Celsius hour per calorie
$^{\circ}\text{C} \cdot \text{s}/\text{kcal}$	degree Celsius second per kilocalorie
$^{\circ}\text{C} \cdot \text{min}/\text{kcal}$	degree Celsius minute per kilocalorie
$^{\circ}\text{C} \cdot \text{h}/\text{kcal}$	degree Celsius hour per kilocalorie
$^{\circ}\text{F} \cdot \text{s}/\text{Btu}$	degree Fahrenheit second per British thermal unit
$^{\circ}\text{F} \cdot \text{h}/\text{Btu}$	degree Fahrenheit hour per British thermal unit
$^{\circ}\text{F}/\text{W}$	degree Fahrenheit per watt
$^{\circ}\text{F}/\text{kW}$	degree Fahrenheit per kilowatt
$^{\circ}\text{F} \cdot \text{h}/\text{J}$	degree Fahrenheit hour per joule
$^{\circ}\text{F} \cdot \text{h}/\text{kJ}$	degree Fahrenheit hour per kilojoule
$^{\circ}\text{K} \cdot \text{h}/\text{J}$	degree Kelvin hour per joule
$^{\circ}\text{K} \cdot \text{h}/\text{kJ}$	degree Kelvin hour per kilojoule
$^{\circ}\text{K} \cdot \text{h}/\text{MJ}$	degree Kelvin hour per megajoule
$^{\circ}\text{K}/\text{W}$	degree Kelvin per watt
$^{\circ}\text{K}/\text{kW}$	degree Kelvin per kilowatt
$^{\circ}\text{K}/\text{MW}$	degree Kelvin per megawatt

Thermal Resistivity

Symbol	Name
$^{\circ}\text{C} \cdot \text{mm}/\text{W}$	degree Celsius millimeter per watt
$^{\circ}\text{C} \cdot \text{cm}/\text{W}$	degree Celsius centimeter per watt
$^{\circ}\text{C} \cdot \text{m}/\text{W}$	degree Celsius meter per watt
$^{\circ}\text{C} \cdot \text{mm}/\text{kW}$	degree Celsius millimeter per kilowatt
$^{\circ}\text{C} \cdot \text{cm}/\text{kW}$	degree Celsius centimeter per kilowatt
$^{\circ}\text{C} \cdot \text{m}/\text{kW}$	degree Celsius meter per kilowatt
$^{\circ}\text{C} \cdot \text{s} \cdot \text{mm}/\text{cal}$	degree Celsius second millimeter per calorie
$^{\circ}\text{C} \cdot \text{s} \cdot \text{cm}/\text{cal}$	degree Celsius second centimeter per calorie
$^{\circ}\text{C} \cdot \text{s} \cdot \text{m}/\text{cal}$	degree Celsius second meter per calorie
$^{\circ}\text{C} \cdot \text{s} \cdot \text{mm}/\text{cal}$	degree Celsius second millimeter per kilocalorie
$^{\circ}\text{C} \cdot \text{s} \cdot \text{cm}/\text{cal}$	degree Celsius second centimeter per kilocalorie
$^{\circ}\text{C} \cdot \text{s} \cdot \text{m}/\text{kcal}$	degree Celsius second meter per kilocalorie
$^{\circ}\text{K} \cdot \text{m}/\text{W}$	degree Kelvin meter per watt
$^{\circ}\text{K} \cdot \text{m}/\text{kW}$	degree Kelvin meter per kilowatt
$^{\circ}\text{K} \cdot \text{m}/\text{MW}$	degree Kelvin meter per megawatt

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Thermal Resistivity (cont.)

$^{\circ}\text{K} \cdot \text{h} \cdot \text{m}/\text{J}$	degree Kelvin hour meter per joule
$^{\circ}\text{K} \cdot \text{h} \cdot \text{m}/\text{kJ}$	degree Kelvin hour meter per kilojoule
$^{\circ}\text{K} \cdot \text{h} \cdot \text{m}/\text{MJ}$	degree Kelvin hour meter per megajoule
$^{\circ}\text{F} \cdot \text{s} \cdot \text{in}/\text{Btu}$	Fahrenheit second inch per British thermal unit
$^{\circ}\text{F} \cdot \text{s} \cdot \text{ft}/\text{Btu}$	Fahrenheit second foot per British thermal unit
$^{\circ}\text{F} \cdot \text{h} \cdot \text{in}/\text{Btu}$	Fahrenheit hour inch per British thermal unit
$^{\circ}\text{F} \cdot \text{h} \cdot \text{ft}/\text{Btu}$	Fahrenheit hour foot per British thermal unit
$^{\circ}\text{F} \cdot \text{in}/\text{W}$	Fahrenheit inch per watt
$^{\circ}\text{F} \cdot \text{ft}/\text{W}$	Fahrenheit foot per watt
$^{\circ}\text{F} \cdot \text{in}/\text{kW}$	Fahrenheit inch per kilowatt
$^{\circ}\text{F} \cdot \text{ft}/\text{kW}$	Fahrenheit foot per kilowatt
$^{\circ}\text{F} \cdot \text{h} \cdot \text{in}/\text{J}$	Fahrenheit hour inch per joule
$^{\circ}\text{F} \cdot \text{h} \cdot \text{ft}/\text{J}$	Fahrenheit hour foot per joule
$^{\circ}\text{F} \cdot \text{h} \cdot \text{in}/\text{kJ}$	Fahrenheit hour inch per kilojoule
$^{\circ}\text{F} \cdot \text{h} \cdot \text{ft}/\text{kJ}$	Fahrenheit hour foot per kilojoule

Time

Symbol	Name
ms	millisecond
microsecond	microsecond
shake	shake
nanosecond	nanosecond
s	second
min	minute
h	hour
d	day
week	week
quarter	quarter
month	month
year	year
decade	decade
century	century

Torque

Symbol	Name
N . mm	newton millimeter
N . cm	newton centimeter
N . m	newton meter
mN . m	millinewton meter
kN . m	kilonewton meter
gf . mm	gram-force millimeter
gf . cm	gram-force centimeter
gf . m	gram-force meter
(avoirdupois ounce-force) . in	avoirdupois ounce-force inch
(avoirdupois ounce-force) . ft	avoirdupois ounce-force foot
(avoirdupois ounce-force) . yd	avoirdupois ounce-force yard
lbf . in	pound-force inch
lbf. ft	pound-force foot
lbf . yd	pound-force yard
kgf . m	kilogram-force meter
dyn . cm	dyne centimeter

Velocity

Symbol	Name
mm/s	millimeter per second
mm/min	millimeter per minute
mm/h	millimeter per hour
cm/s	centimeter per second
cm/min	centimeter per minute
cm/h	centimeter per hour
m/s	meter per second
m/min	meter per minute
m/h	meter per hour
m/d	meter per day
km/s	kilometer per second
km/min	kilometer per minute
km/h	kilometer per hour
km/d	kilometer per day

(...continue in next page)

Velocity (cont.)

Symbol	Name
in/s	inch per second
in/min	inch per minute
in/h	inch per hour
ft/s	foot per second
ft/min	foot per minute
ft/h	foot per hour
ft/d	foot per day
yd/s	yard per second
yd/min	yard per minute
yd/h	yard per hour
mi/s	mile per second
mi/min	mile per minute
mi/h	mile per hour
mi/d	mile per day

Dynamic Viscosity

Symbol	Name
s . Pa	second pascal
min . Pa	minute pascal
h . Pa	hour pascal
s . kPa	second kilopascal
min . kPa	minute kilopascal
h . kPa	hour kilopascal
s . dyn/cm ²	second dyne per square centimeter
s . N/mm ²	second newton per square millimeter
s . N/cm ²	second newton per square centimeter
s . N/m ²	second newton per square meter
h . N/mm ²	hour newton per square millimeter
h . N/cm ²	hour newton per square centimeter
h . N/m ²	hour newton per square meter
g/(cm . s)	gram per centimeter second
kg/(m . s)	kilogram per meter second
kg/(m . h)	kilogram per meter hour
poise	poise
cP	centipoise

(...continue in next page)

Dynamic Viscosity (cont.)

Symbol	Name
s . kgf/mm ²	second kilogram force per square millimeter
s . kgf/cm ²	second kilogram force per square centimeter
s . kgf/m ²	second kilogram force per square meter
h . kgf/mm ²	hour kilogram force per square millimeter
h . kgf/cm ²	hour kilogram force per square centimeter
h . kgf/m ²	hour kilogram force per square meter
s . poundal/in ²	second poundal per square inch
s . poundal/ft ²	second poundal per square foot
h . poundal/in ²	hour poundal per square inch
h . poundal/ft ²	hour poundal per square foot
s . lbf/in ²	second pound force per square inch
s . lbf/ft ²	second pound force per square foot
h . lbf/in ²	hour pound force per square inch
h . lbf/ft ²	hour pound force per square foot
slug/(in . s)	slug per inch second
slug/(ft . s)	slug per foot second
slug/(in . h)	slug per inch hour
slug/(ft . h)	slug per foot hour

Kinematic Viscosity

Symbol	Name
mm^2/s	square millimeter per second
mm^2/min	square millimeter per minute
mm^2/h	square millimeter per hour
cm^2/s	square centimeter per second
cm^2/min	square centimeter per minute
cm^2/h	square centimeter per hour
cSt	centistokes
m^2/s	square meter per second
m^2/h	square meter per hour
in^2/s	square inch per second
in^2/h	square inch per hour
ft^2/s	square foot per second
ft^2/h	square foot per hour
St	stokes

Volume

Symbol	Name
mm^3	cubic millimeter
cm^3	cubic centimeter
mL	milliliter
L	liter
m^3	cubic meter
km^3	cubic kilometer
in^3	cubic inch
ft^3	cubic foot
yd^3	cubic yard
mi^3	cubic mile
acre . in	acre inch
acre . ft	acre foot
U.K. barrel	U.K. barrel
U.S. barrel petroleum	U.S. barrel petroleum
U.K. bushel	U.K. bushel
U.S. bushel	U.S. bushel
cord	cord

(...continue in next page)

Volume (cont.)

Symbol	Name
cup	cup
U.K. gal	U.K. gallon
U.S. gal	U.S. gallon
U.K. gill	U.K. gill
U.S. gill	U.S. gill
U.K. fluid ounce	U.K. fluid ounce
U.S. fluid ounce	U.S. fluid ounce
U.K. peck	U.K. peck
U.S. peck	U.S. peck
U.S. dry pint	U.S. dry pint
U.S. liquid pint	U.S. liquid pint
U.S. pipe	U.S. pipe
U.S. dry quart	U.S. dry quart
U.S. liquid quart	U.S. liquid quart
stere	stere
table spoon	table spoon
tea spoon	teaspoon

Volume Divided by Energy

Symbol	Name
mm^3/cal	cubic millimeter per calorie
cm^3/cal	cubic centimeter per calorie
mL/cal	milliliter per calorie
L/cal	liter per calorie
m^3/cal	cubic meter per calorie
cm^3/kcal	cubic centimeter per kilocalorie
mL/kcal	milliliter per kilocalorie
L/kcal	liter per kilocalorie
m^3/kcal	cubic meter per kilocalorie
$\text{L}/(\text{hp} \cdot \text{h})$	liter per horsepower hour
$\text{m}^3/(\text{hp} \cdot \text{h})$	cubic meter per horsepower hour
U.K. gal/cal	U.K. gallon per calorie
U.K. gal/kcal	U.K. gallon per kilocalorie
U.K. gal/Btu	U.K. gallon per British thermal unit
U.K. gal/(hp · h)	U.K. gallon per horsepower hour
U.S. gal/cal	U.S. gallon per calorie
U.S. gal/kcal	U.S. gallon per kilocalorie

(...continue in next page)

Volume Divided by Energy (cont.)

Symbol	Name
mm^3/J	cubic millimeter per joule
cm^3/J	cubic centimeter per joule
mL/J	milliliter per joule
L/J	liter per joule
m^3/J	cubic meter per joule
cm^3/kJ	cubic centimeter per kilojoule
L/kJ	liter per kilojoule
m^3/kJ	cubic meter per kilojoule
m^3/MJ	cubic meter per megajoule
$\text{cm}^3/(\text{W} \cdot \text{h})$	cubic centimeter per watt hour
$\text{L}/(\text{W} \cdot \text{h})$	liter per watt hour
$\text{m}^3/(\text{W} \cdot \text{h})$	cubic meter per watt hour
$\text{m}^3/(\text{kW} \cdot \text{h})$	cubic meter per kilowatt hour
in^3/Btu	cubic inch per British thermal unit
ft^3/Btu	cubic foot per British thermal unit
U.S. gal/Btu	U.S. gallon per British thermal unit
U.S. gal/(hp · h)	U.S. gallon per metric horsepower hour

